



ZEV Adoption and Air Quality in Southern California:

A Post-COVID Panel Analysis of NO₂ and PM_{2.5}

Across 20 Cities (2021–2024)

by

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Abstract

This study examines whether localized ZEV adoption rates correlate with the persistence of NO₂ and PM_{2.5} reductions across 20 Southern California cities (2021-2024). Using EPA AirData, California Energy Commission vehicle records, Census ACS income, and CalEnviroScreen 4.0 scores, I estimate four OLS panel models with year fixed effects. To address severe collinearity, I apply the Frisch-Waugh-Lovell theorem to residualize ZEV adoption and CES scores with respect to pollutants across all specifications. ZEV adoption is not associated with air quality improvement, as cities growing their ZEV fleets fastest are also the most traffic-intensive. The equity interaction is not significant, indicating no detectable difference in the ZEV effect across vulnerability quartiles. Residual analysis confirms that high-burden communities are systematically more polluted than predicted, pointing to heavy-duty freight as the mechanism that sustains environmental inequity beyond the reach of light-duty electrification policy.

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1. Introduction

Breathing clean air in Southern California has never been equally distributed. Long before COVID-19, communities in the Inland Empire recorded some of the worst NO_2 and particulate matter levels in the U.S., and all of this is based on design. Decades of highway routing, industrial siting, and warehouse expansion concentrated pollution in low-income communities of color while cleaner neighborhoods remained insulated. Tessum et al. (2019) quantified this disparity at the national level, finding that Black and Hispanic Americans bear 56 and 63% excess $\text{PM}_{2.5}$ exposure relative to what their own consumption causes. Non-Hispanic white Americans experience around 17% less pollution than they generate, but the pandemic briefly interrupted that pattern. Many studies show there was an overall drop in these pollutants, yet even after those drops, the least White census tracts still experience roughly 1.5 times higher NO_2 than the most White census tracts (Kerr 2021). As vehicle activity resumed post-pandemic, one of the questions that arises from this is whether accelerating the zero-emission vehicle program (in California) could sustain any portion of those gains on a durable and equitable basis.

This study addresses three primary research questions. Do localized zero-emission vehicle (ZEV) adoption rates correlate with reductions in nitrogen dioxide (NO_2) and fine particulate matter ($\text{PM}_{2.5}$) concentrations across 20 Southern California cities between 2021 and 2024? Does ZEV adoption in a given year predict air quality outcomes in the subsequent year? And are any observed air quality benefits distributed equitably across communities stratified by vulnerability quartiles as defined by CalEnviroScreen?

2. Literature Review

COVID lockdowns produced a clear shift in ambient air quality as a result of vehicle activity. Saha et al. (2022) documented that NO_2 and $\text{PM}_{2.5}$ declined by more than 60% in many global megacities during these lockdowns. In Southern California, Parker et al. (2020)

confirmed measurable improvements associated with traffic reductions. Kerr et al. (2021) showed that while the largest drops occurred in low-income and minority neighborhoods, racial NO_2 disparities persisted because heavy-duty trucking continued unabated. They conclude that targeting NO_x from heavy-duty diesel is the most effective strategy for reducing these disparities nationwide, not passenger vehicle electrification.

Tessum et al. (2019) explore the social dimension of this research question at a broader level, finding that non-Hispanic white Americans experience about 17% less $\text{PM}_{2.5}$ than is caused by their consumption, while Black and Hispanic Americans bear 56 to 63% excess exposure, respectively. $\text{PM}_{2.5}$ exposures declined 50% from 2002 to 2015 for all groups, but pollution inequity remained persistently high.

Garcia et al. (2023) provided the first practical evidence in California linking ZEV adoption to improved air quality and health, finding a 3.2% decrease in asthma-related emergency department visits for every increase of 20 ZEVs per 1,000 people. Although the association with NO_2 was non-significant, the directional effect remained negative. Their adoption gap finding, lower-attainment zip codes grew ZEV fleets at 1/5 the rate of high-attainment zip codes, foreshows the equity question this study tests directly. Wang et al. (2026) document that affluent urban households still account roughly 64% of all EV owners despite growing demographic diversity in the EV owner population.

Together, this literature generates three hypotheses: CalEnviroScreen burden should persistently predict air quality even after controlling for income and ZEV adoption; ZEV adoption at current penetration rates may not produce detectable ambient improvements; and the equity distribution of any ZEV effect likely follows the adoption gap, with high-burden communities capturing fewer benefits.

3. Data and Methods

3.1. Sample and Data Sources

The study covers 20 Southern California cities with continuous EPA monitor coverage for both NO_2 and $\text{PM}_{2.5}$ across all four observation years (2021–2024), yielding a balanced panel of 80 city-year observations. The balanced-panel requirement was reduced by 2/3 of candidate cities but ensures that every regression observation is methodologically equivalent.

Air quality data come from the EPA AirData AQS annual summary files. $\text{PM}_{2.5}$ uses parameter code 88101, filtered to the Annual pollutant standard with the “Quarterly Means of Daily Means” metric, the EPA’s method for computing a valid annual mean by averaging those four values into a single annual figure. Only monitors with at least 75% percent observation completeness are retained, consistent with EPA’s own validity. For NO_2 , it uses parameter code 42602 and is filtered to the “ NO_2 Annual 1971” standard, which isolates the long-term annual average records and excludes monitors reporting under the 1-hour peak standard. The same 75% completeness threshold applies. ZEV registration data comes from the California Energy Commission Vehicle Population dataset, aggregated from ZIP codes to cities using SimpleMaps. Census ACS 2020 5-Year estimates provide tract-level median household income, aggregated to the city level via HUD USPS ZIP-to-tract crosswalk weights. CalEnviroScreen 4.0 provides tract-level composite burden scores averaged to the city level by ZIP code matching. All control and moderation variables are fixed at 2020 pre-period values to avoid endogeneity.

Before running the regressions, variance inflation factor analysis on the raw variables confirms severe multicollinearity. ZEV adoption rate and CES score have VIF values of around 12.4 and 8.7 in the joint model, meaning the variables are so correlated with each other that the regression cannot separate their individual contributions. The root cause is a shared wealth gradient, wealthier cities tend to have both higher ZEV adoption and lower environmental burden simultaneously. To resolve this, I apply the Frisch–Waugh–Lovell

theorem, which states that the coefficient on a variable in a multiple regression is equivalent to the coefficient you get if you first partial out the other regressors from both that variable and the outcome. In practice, I regress `ZEV_rate` on income and fleet size, take the residuals, and use those residuals in the main model instead. The same applies to the `CES_score`. After residualization, VIF for both variables drops to approximately 1.1, and the correlation between `ZEV_rate_resid` and `CES_score_resid` falls near zero compared to the moderate negative correlation in the raw variables.

3.2. Key Variables

Dependent variables: Annual mean $\text{PM}_{2.5}$ concentration ($\mu\text{g}/\text{m}^3$) and annual mean NO_2 concentration (ppb).

Independent variables: `ZEV_rate` is the share of ZEV vehicles (BEV + PHEV + FCEV) in the total registered fleet per city-year. `ZEV_rate_lag` is the prior year’s adoption rate, used in Models 3 and 4. The 2021 lag is missing because the prior year falls in the baseline, so the lagged models runs on 2022–2024 only ($N = 60$)

Moderating variable: `CES_score` is the tract-level CalEnviroScreen composite score averaged to the city level, fixed at 2020 values. Higher scores indicate greater environmental burden and population vulnerability.

Control variables: `income_10k` is median household income in \$10,000 units, fixed at 2020 ACS values. `log_total_vehicles` is the natural log of total registered vehicles per city-year, controlling for city scale.

3.3. Regression Models

Four OLS models are estimated separately for $\text{PM}_{2.5}$ and NO_2 with city-clustered standard errors and year fixed effects throughout. The reference year is 2021.

Model 1 (Baseline) regresses air quality on income and year dummies alone, establishing the baseline socioeconomic gradient:

$$AQ_{i,t} = \alpha + \beta_1 \cdot \text{income}_{10k,i} + \gamma_t + \varepsilon_{i,t} \quad (1)$$

where $AQ_{i,t}$ is the annual mean concentration of NO_2 or $\text{PM}_{2.5}$ in city i in year t ; α is the intercept; β_1 is the coefficient of median household income (in \$10,000 units); γ_t is a year fixed effect; and $\varepsilon_{i,t}$ is the error term.

Model 2 (Policy) adds the residualized ZEV rate, log fleet size, and residualized CES score. The β on ZEV is expected to be negative (more ZEVs, less pollution) and the β on CES is expected to be positive (more burden, more pollution):

$$AQ_{i,t} = \alpha + \beta_1 \cdot \text{income}_i + \beta_2 \cdot \log(\text{fleet}_{i,t}) + \beta_3 \cdot \widehat{ZEV}_{i,t} + \beta_4 \cdot \widehat{CES}_i + \gamma_t + \varepsilon_{i,t} \quad (2)$$

where $\widehat{ZEV}_{i,t}$ is the residualized ZEV adoption rate (ZEV share unexplained by income and fleet size); $\widehat{CES}_i$ is the residualized CalEnviroScreen score (burden unexplained by income); $\log(\text{fleet}_{i,t})$ is the natural log of total registered vehicles; and all other terms are as defined above.

Model 3 (Persistence) replaces the contemporaneous ZEV residual with its one-year lag. A significant negative β would indicate that ZEV adoption precedes, rather than coincide with cleaner air:

$$AQ_{i,t} = \alpha + \beta_1 \cdot \text{income}_i + \beta_2 \cdot \log(\text{fleet}_{i,t}) + \beta_3 \cdot \widehat{ZEV}_{i,t-1} + \beta_4 \cdot \widehat{CES}_i + \gamma_t + \varepsilon_{i,t} \quad (3)$$

where $\widehat{ZEV}_{i,t-1}$ is the residualized ZEV adoption rate lagged one year (prior year's ZEV share unexplained by income and fleet size). All other terms are as defined in Model 2.

Model 4 (Equity) adds an interaction between the lagged ZEV residual and the CES residual. A positive $\beta_5 > 0$ signals an equity gap; a negative $\beta_5 < 0$ signals an equity

gain; a coefficient near zero indicates uniform effects across vulnerability levels:

$$AQ_{i,t} = \alpha + \beta_1 \cdot \text{income}_i + \beta_2 \cdot \log(\text{fleet}_{i,t}) + \beta_3 \cdot \widehat{ZEV}_{i,t-1} \\ + \beta_4 \cdot \widehat{CES}_i + \beta_5 \cdot (\widehat{ZEV}_{i,t-1} \times \widehat{CES}_i) + \gamma_t + \varepsilon_{i,t} \quad (4)$$

where β_5 is the coefficient on the interaction between the lagged residualized ZEV rate and the residualized CES score, capturing whether the ZEV–air quality relationship differs across vulnerability levels. All other terms are as defined above.

Following the regressions, a residual analysis plots city-level Model 3 residuals against CES scores from a restricted model that excludes CES itself, testing whether unmeasured burden- related sources generate a pollution premium that the model cannot explain.

4. Results

Sample

The 20 verified cities are: Anaheim, Banning, Compton, El Cajon, Fontana, Lancaster, Lompoc, Long Beach, Los Angeles, Mira Loma, Ontario, Palm Springs, Pasadena, Pico Rivera, Reseda, Rubidoux, San Bernardino, San Diego, Simi Valley, and Victorville. CalEnviroScreen scores range from 15.4 (Palm Springs) to 61.1 (Compton). ZEV adoption rates range from 1.0% (Compton) to 7.3% (Pasadena). The 2023 dip in $\text{PM}_{2.5}$ visible across nearly all cities reflects the unusually wet atmospheric river year; year fixed effects absorb this shock.

Regression Results

Table 1: OLS Results — PM_{2.5} Annual Mean ($\mu\text{g}/\text{m}^3$)

Variable	Model 1	Model 2	Model 3	Model 4
income_10k	0.046	0.046	−0.015	0.052
ZEV_rate_resid	—	47.34***	—	—
ZEV_rate_lag_resid	—	—	50.63*	66.99**
CES_score_resid	—	0.216***	0.204***	0.215***
ZEV $\times$ CES	—	—	—	1.49
N	80	80	60	60
R^2	0.061	0.605	0.601	0.584
Adj. R^2	0.011	0.572	0.556	0.537

* $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$. *City-clustered SEs. Year dummies included; reference year 2021. Models 3–4 run on 2022–2024 only ($N = 60$). `log_total_vehicles` included in Models 2–4 but omitted for brevity.*

Table 2: OLS Results — NO₂ Annual Mean (ppb)

Variable	Model 1	Model 2	Model 3	Model 4
income_10k	−0.034	−0.034	−0.393	−0.105
ZEV_rate_resid	—	99.17*	—	—
ZEV_rate_lag_resid	—	—	109.61*	152.62*
CES_score_resid	—	0.421***	0.388***	0.426***
ZEV × CES	—	—	—	3.48
N	80	80	60	60
R^2	0.014	0.426	0.521	0.446
Adj. R^2	−0.039	0.379	0.467	0.383

* $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$. *City-clustered SEs. Year dummies included; reference year 2021. Models 3–4 run on 2022–2024 only ($N = 60$).*

Key Findings

Income (Model 1). The income coefficient is close to zero and not statistically significant for either pollutant ($R^2 = 0.061$ for PM_{2.5}; 0.014 for NO₂). This likely does not indicate that income is unimportant for air quality, but that within this sample, the income range is too compressed relative to the statewide distribution to identify income effects.

CalEnviroScreen (Models 2–4). CES_score_resid is positive and highly significant across every model and both pollutants. For PM_{2.5}, the coefficient ranges from 0.20 to 0.22 $\mu\text{g}/\text{m}^3$ per unit; for NO₂, it ranges from 0.39 to 0.43 ppb per unit. This stability is the study’s strongest result: pre-existing environmental burden predicts current air quality even after controlling for income, fleet size, ZEV adoption, and year trends, consistent with Kerr et al. (2021) and Tessum et al. (2019).

ZEV coefficient (Models 2 and 3). ZEV_rate_resid is positive and statistically significant for both PM_{2.5} (47.34, $p < 0.01$) and NO₂ (99.17, $p < 0.10$) in Model 2. The lagged

specification in Model 3 produces similar positive and significant coefficients ($PM_{2.5}$: 50.63, $p < 0.10$; NO_2 : 109.61, $p < 0.10$). After resolving collinearity, this is a genuine empirical result: cities with unexpectedly high ZEV adoption given their income and fleet size are associated with worse, not better, air quality. The lagged coefficient is also larger, which rules out the idea that ZEVs need more time, if delayed benefits were accumulating, the lag should be smaller.

Residual Analysis

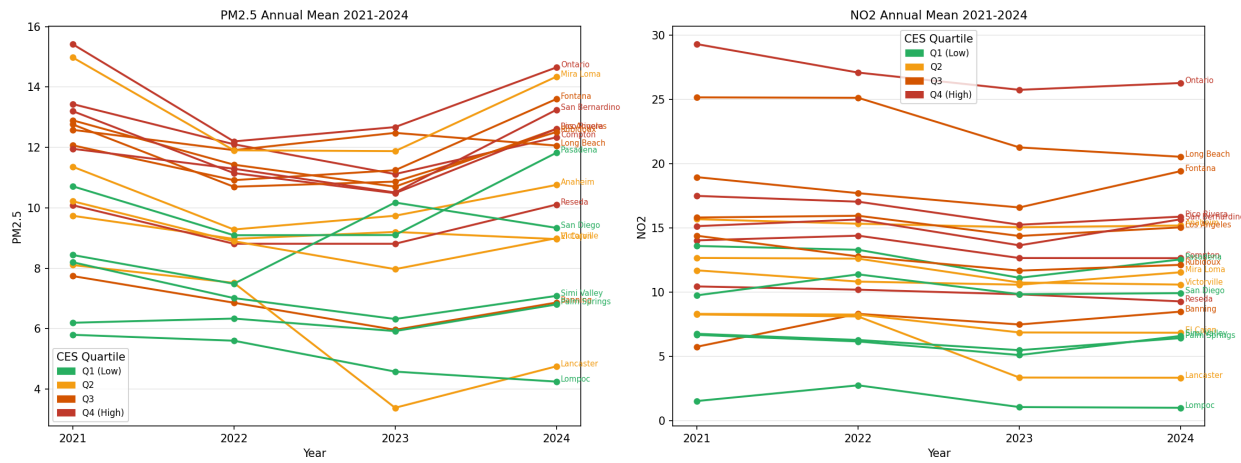


Figure 1: $PM_{2.5}$ and NO_2 trend lines 2021–2024, colored by CES quartiles.

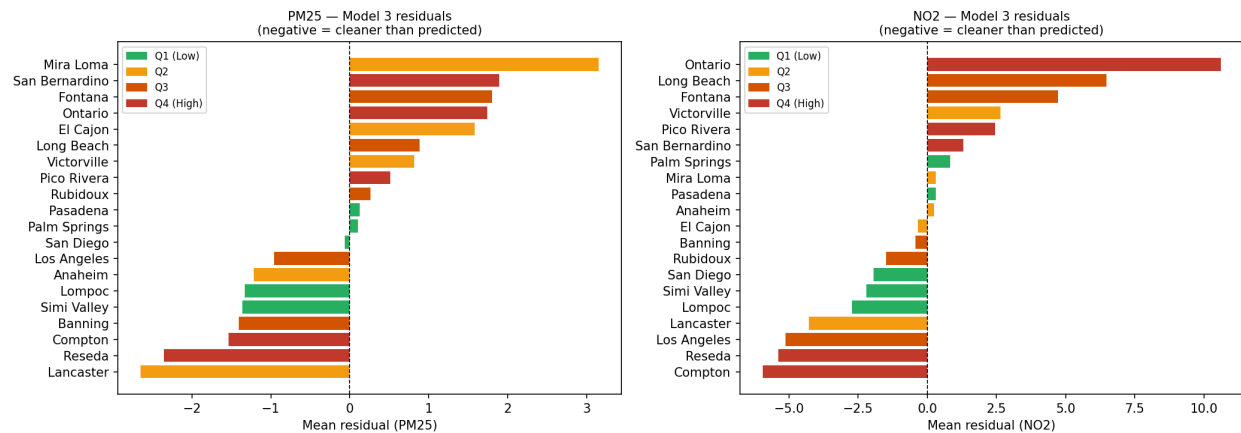


Figure 2: Model 3 residual bar charts by city, colored by CES quartile. Negative = cleaner than predicted

achieve net NO_x reductions.

The robustness of the CES coefficient is the most important policy-related finding. Environmental burden explains a large share of cross-city air quality variation independently of any concurrent policy variables. This is consistent with Tessum et al.’s (2019) finding that pollution inequality is structural, and with Kerr et al.’s (2021) finding that even a near-elimination of passenger traffic failed to close racial NO_2 disparities because heavy-duty trucking persists. The residual analysis replicates that mechanism at the city level: Ontario, Mira Loma, and Fontana are systematically more polluted than their income and ZEV adoption would predict.

The non-significant equity interaction means there is no detectable differential in how ZEV adoption affects air quality across high-versus low-burden communities. High-burden communities are not benefiting more, but they are also not benefiting less. The structural inequity captured by the CES coefficient persists regardless of local ZEV penetration and likely will continue to unless policy directly addresses freight and industrial sources that light-duty electrification does not reach. Rule 2305 (the WAIRE program) is designed for this, but its mitigation fee structure means compliance does not guarantee emission reductions in the specific communities bearing the burden.

6. Conclusion

This study estimated the relationship between localized ZEV adoption and air quality persistence across 20 Southern California cities from 2021 to 2024, and found three main results. CalEnviroScreen burden is the dominant predictor of both NO_2 and $\text{PM}_{2.5}$, robust across all model specifications and consistent with the structural inequality literature. Second, ZEV adoption at current penetration rates is not associated with air quality improvement, and the lagged persistence test confirms that no delayed benefit is accumulating over the one-year horizon studied. Third, there is no detectable equity differential in ZEV air quality effects.

These results support Kerr et al.’s (2021) conclusion that strategies targeting heavy-duty diesel emissions are likely more effective at reducing air pollution disparities than passenger vehicle electrification alone. They are also consistent with Wang et al.’s (2026) concern that even if ZEVs eventually produce ambient air benefits, those benefits will accumulate in communities that need them least, given that affluent urban households still account for roughly 64% of EV owners.

Three extensions are priorities for future work: extending the panel through 2025–2026 as AQS data become available; incorporating Rule 2305 compliance data to test whether warehouse-sector clean truck deployment provides benefits beyond light-duty electrification; and deploying a tract-level analysis to capture intra-city variation in ZEV adoption and exposure.

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